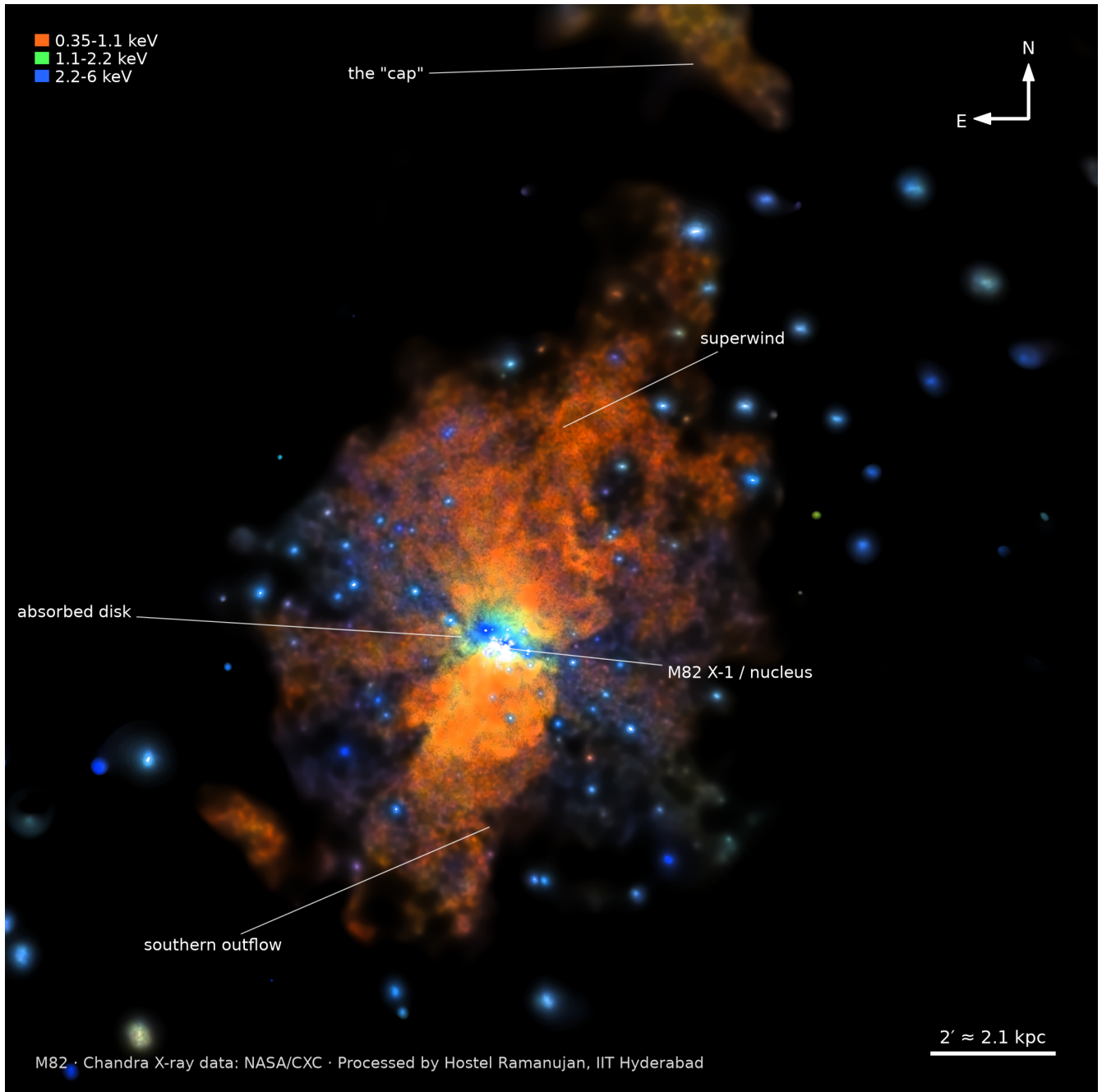


M82 in X-rays

Cepheid Astro-Photography Competition, IIT Hyderabad · stacking report

Hostel: **Ramanujan** · Team: Piyush Anand, Darshanraj Pattanaik · 30 September 2026

Code: github.com/Anandpiyush21/Astrophotography_Milan_2026



Final image, annotated. $18' \times 18'$, north up, east left, $0.492''$ per pixel. Orange is 0.35–1.1 keV, green 1.1–2.2 keV, blue 2.2–6 keV. The submitted file is the clean version, `M82_Chandra_wide_Ramanujan.png` (a 16-bit TIFF master is also kept). Data: NASA/CXC.

Summary

The data we were given are not a set of exposures to align and average. They are seven Chandra X-ray images of M82, one per energy band, already on the same pixel grid. Most pixels hold zero or one photon. For us, stacking meant two things: combining the seven bands into one image, and dealing with the photon noise without blurring the point sources.

We did this in Python in five steps: estimate the missing exposure map, smooth each pixel only as much as it needs, build brightness from all bands and colour from three energy groups, then stretch and finish. In the faint wind the signal-to-noise per pixel went from 0.64 to 8.4, while point sources stayed as sharp as in the raw data (8.2 px against 8.8 px). An ordinary 8 px blur gets 7.6 and spreads the same source to 14.6 px.

The whole process is scripted. A single command rebuilds the submitted image and this report from the raw FITS files (Section 10). No manual painting, masking or retouching was used.

1. M82

M82 is a starburst galaxy about 3.6 Mpc (12 million light-years) away, seen nearly edge-on. A close pass by M81 set off very fast star formation in its centre. The supernovae that followed heat gas to millions of degrees and push it out above and below the disk. This outflow, the superwind, is bright in soft X-rays and is the orange structure in our image. The image also shows:

- point sources, mostly X-ray binaries, including the very bright M82 X-1 and, in the nucleus, the ultraluminous pulsar M82 X-2;
- the disk itself, which shows up blue because its gas and dust absorb the low-energy X-rays;
- the "cap", a faint cloud about 11 kpc north of the galaxy where the wind runs into surrounding gas.

2. The data

Band	Photons	Share	Net photons (r < 4.9')	Background (cts/px)	Max pixel	Pixels with a photon	Main content
350-500 eV	42,314	2.2%	18,870	0.0043	12	0.43%	O VII / O VIII lines, coolest wind gas
500-700 eV	142,811	7.3%	109,647	0.0053	30	1.23%	O and Fe-L lines, soft thermal wind
700-1100 eV	642,433	32.9%	576,205	0.0102	368	2.56%	Fe-L and Ne lines, brightest band
1100-1600 eV	409,454	20.9%	346,013	0.0106	637	1.62%	Mg and Ne lines, warm gas
1600-2200 eV	264,866	13.5%	190,279	0.0132	468	1.32%	Si lines, hot gas near the starburst
2200-2800 eV	115,446	5.9%	64,890	0.0091	229	0.76%	S lines and hard continuum
2800-6000 eV	337,617	17.3%	139,816	0.0362	3070	2.41%	hard continuum: X-ray binaries, absorbed nucleus
Total	1,954,941	100%					

- Each file is 3072 × 3072 pixels of 32-bit floats holding whole photon counts, at 0.492" per pixel (CDELTA = 0.000136667 deg), Chandra's native pixel size. The projection is tangent-plane (TAN), centred on RA 148.967°, Dec 69.6802°.
- Even after summing all seven bands, 87% of the pixels inside the observed area hold no photon at all. The noise is Poisson (shot) noise, not the Gaussian noise of a camera.
- The field is a mosaic of several pointings taken at different angles, and no exposure maps were included. Because of that, the detector chip edges show up as steps in the background (Fig. 2a).

- The brightest pixel (x=1543, y=1525) has 3257 photons, 94% of them above 2.8 keV. This is M82 X-1, and it suffers from pile-up. Right next to it is a diagonal strip of 974 pixels with zero counts in every band. From the pixels around it, the strip should hold about 12 photons per pixel, 11,861 in total. For Poisson counts the chance of getting none is $e^{-11,861}$, so this is not real sky. It is most likely a column removed during the original processing.

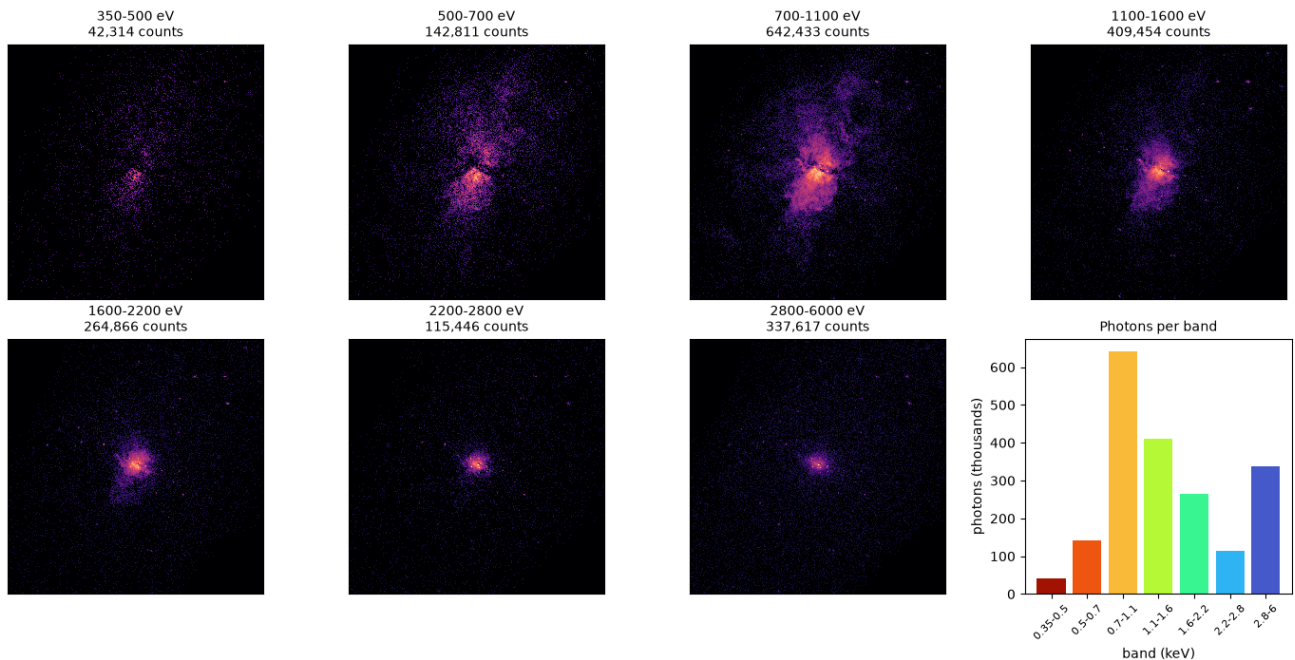


Fig. 1. The seven raw bands (2×2 binned, asinh). The wind dominates below 1.1 keV, and point sources and the nucleus dominate above 2.2 keV.

3. Alignment and stacking

Alignment check. We compared the coordinate keys (CTYPE, CRVAL, CRPIX, CDELT) of all seven headers key by key, and they are identical. Identical headers do not prove the pixels line up, so each band was also cross-correlated against the all-band sum (phase correlation, central 1000 × 1000 px, 3 px pre-blur, 1/20 px precision). The largest offset was 0.00 px, far below Chandra's 0.5" resolution. We therefore did not resample any band, which avoids interpolation blur.

Why a plain sum. The pixels are photon counts. A sum of Poisson counts is again Poisson, and every photon gets the same weight. A mean has the same S/N, and a median throws information away, because most pixels are zero. So the brightness layer is the plain sum of the seven bands (1,954,941 photons), and the bands are also kept apart for colour.

4. Step 1: exposure map and background (01_prep.py)

Counts are flux times exposure, so parts of the mosaic covered by more pointings look brighter. Without a correction, these steps turn into visible seams once the image is smoothed.

The exposure had to be estimated from the data. In the 2.8–6 keV band, away from sources, most events come from the detector's particle background, which builds up at a steady rate per second of exposure. There the galaxy is also compact and easy to mask. So we:

1. masked point sources and a 480 px circle around the nucleus;
2. binned the hard band to 16×16 px and took a median over 5×5 bins, which keeps the chip edges sharp;

3. filled the masked area with a Laplace (harmonic) fill and scaled the result to a median of 1 (Fig. 2c);
4. measured the background level of each band only beyond 1200 px from the centre, so the wind is not counted as background.

The spread of the sky level between 64×64 px blocks fell from 64% to 30%, and the seams are gone (Fig. 2d). Pixels in the zero-count strip near X-1 were filled in from their neighbours.

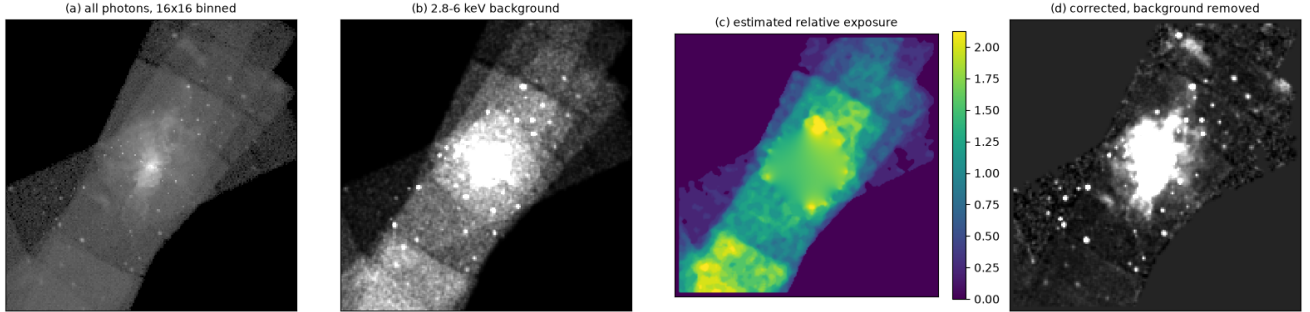


Fig. 2. (a) All photons, showing the mosaic. (b) The hard-band background. (c) Estimated exposure. (d) After correction.

5. Step 2: adaptive smoothing (02_adaptive_smooth.py)

No single blur works for both the point sources and the faint wind. We used the approach of CIAO's `csmooth` (Ebeling et al. 2006). For each pixel we try Gaussians from 0.5 to 32 px and keep the smallest one where the signal is significant above the background:

$$(G * D - G * B) / \sqrt{(\max(G * D, G * B) \cdot \Sigma g^2)} \geq \tau$$

Here D is the photon image, B is the background (rate × exposure), and Σg^2 turns the Poisson variance into the variance of the smoothed value. Between two kernel sizes we interpolate, so no edges appear where the size changes. Flux is $G * (D - B) / G * E$, with counts and exposure blurred the same way, which also takes care of the mosaic edges. All blurs are done with FFTs, about 10 s per band for the full field.

On the first run the wind looked grainy. Some pixels passed the test at tiny kernels purely because of upward noise fluctuations. A 5×5 median on the map of chosen sizes removes these single pixels. Real sources cover several pixels, so they are not affected.

The brightness layer uses $\tau = 6$ on the sum of all bands. The colour layer uses a stricter $\tau = 15$, and the same kernel map is applied to every band, so the colour ratios always come from equally smoothed data.

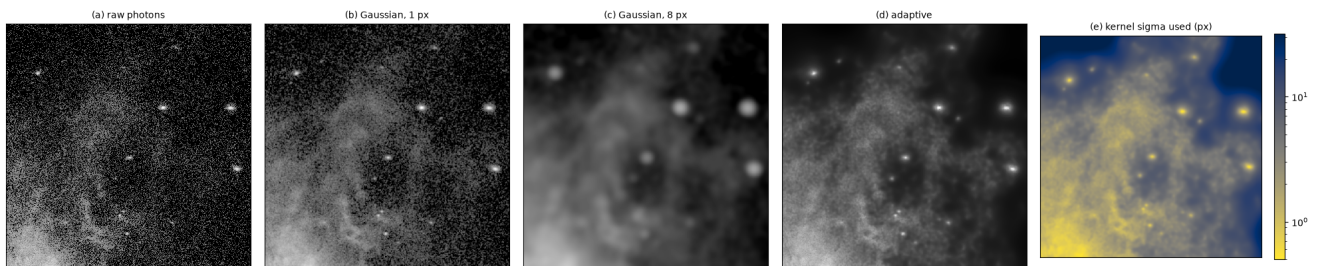


Fig. 3. The north-west part of the wind. Fixed blurs are either noisy (b) or smear the sources (c). The adaptive version (d) is smooth in the gas and sharp on the sources. (e) shows the kernel size used at each pixel.

Method	Sky RMS	Wind S/N per pixel	Point source FWHM (px)
raw	0.439	0.64	8.8
Gaussian 1 px	0.128	2.21	8.0
Gaussian 2 px	0.070	4.03	8.8
Gaussian 4 px	0.046	6.18	11.2
Gaussian 8 px	0.038	7.59	14.6
adaptive	0.034	8.41	8.2

The FWHM is measured on the source at (1863, 2011), about 5' off-axis, where the telescope's point spread function is already a few pixels wide.

6. Step 3: colour (`03_compose.py`)

Brightness is the sum of all seven bands. Colour comes from three groups: 0.35–1.1 keV (the cooler wind gas), 1.1–2.2 keV, and 2.2–6 keV (binaries and the absorbed nucleus). We show these as orange, green and blue. We tried pure red, green and blue first, but the wind came out a flat, harsh red.

The colour layer is blurred another 5 px. The eye notices colour detail much less than brightness detail, the same idea as LRGB imaging. The whole galaxy is balanced to neutral, so a colour just means softer or harder than the average. Following Lupton et al. (2004), only the colour ratios are applied to the stretched brightness, so stretching does not shift the hues.

Reading the colours. Orange is soft, thermal emission from oxygen and iron lines in the wind. Toward the centre the colour moves through yellow and green to blue as the gas gets hotter and denser, and as the dusty disk absorbs the softest photons. The blue haze along the disk is real hard emission from unresolved X-ray binaries. The compact blue and white points are hard-spectrum binaries, including M82 X-1, and background quasars seen through the galaxy.

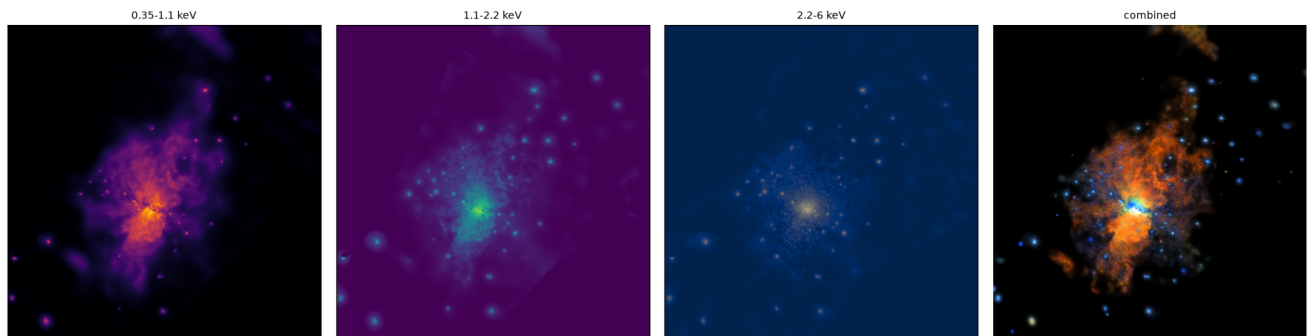


Fig. 4. The three energy groups and the combined image.

7. Step 4: stretch and finishing

1. Black point at 0.5σ of the sky noise. White point at the 99.995 percentile.
2. asinh stretch, softening 1.1σ for the wide image and 12.0σ for the core, so both the faint gas and the bright centre show.
3. Non-local-means denoising, applied only where the kernel was larger than 2 px, i.e. the gas and not the sources.
4. Local contrast (unsharp mask at $\sigma = 40$ px, amount 0.35) to bring out the wind's filaments.
5. Sharpening ($\sigma = 1.2$ px, amount 0.4), only where the kernel was below 2 px. Sharpening the noisy areas would only have made up structure.

6. Saturation $\times 1.45$, eased in with brightness so faint noisy areas stay neutral. The brightest parts fade to white.
7. A 40 px fade at the mosaic edge, and export with north up as 16-bit TIFF, PNG and JPEG.

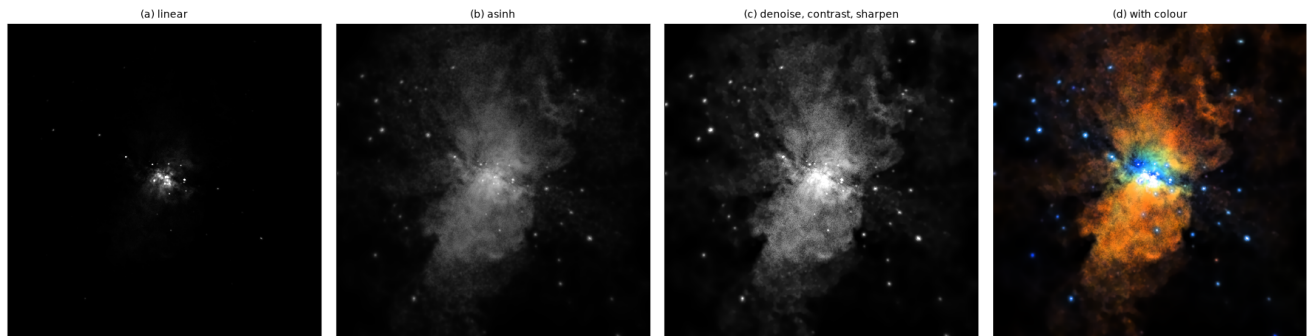
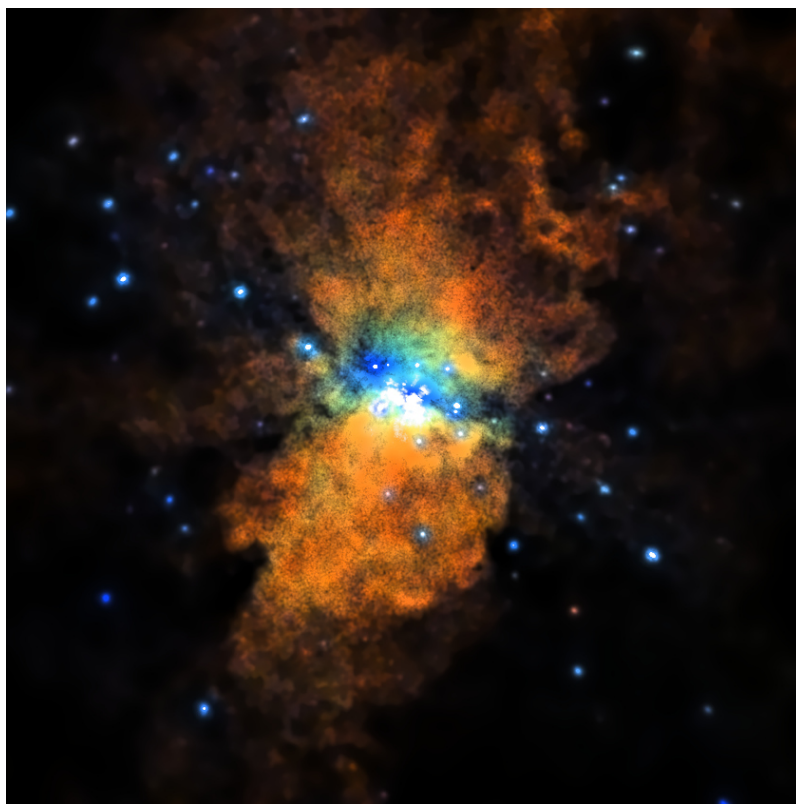


Fig. 5. The core at each stage.



Close-up of the centre (6.9'). The blue absorbed disk crosses the white core, with the wind rising on both sides.

8. Problems along the way

What we saw	Cause	Fix
Exposure map about 3× too high near the galaxy	Our first estimate used all bands, and the galaxy's soft halo leaked into the "background"	Used only the 2.8–6 keV band and masked a wider area
Streaks radiating from the centre of the exposure map	The way the masked hole was filled	Switched to a Laplace fill
A black spot, later a checkerboard, at X-1	The zero-count strip. Setting its exposure to zero made the flux unstable at small kernels	Filled the counts in from neighbouring pixels instead
Grainy wind	Noise pixels passing at tiny kernels, made worse by a minimum filter we had used	Median on the kernel map, τ raised from 4 to 6, denoising in the gas
Speckled colour and a purple haze	Noisy high-energy background clipped at zero	Stricter colour threshold, extra colour blur, less saturation in faint areas

9. Limitations

- The exposure map is an estimate. The observation files needed to compute it properly were not provided.
- We did not deconvolve. Sources far from the centre look larger because Chandra's resolution drops off-axis.
- M82 X-1 is piled up, so its brightness and colour are not reliable.
- X-rays have no colour. Ours are a mapping from energy, with orange for low and blue for high.

10. Code and reproduction

All code is on GitHub at github.com/Anandpiyush21/Astrophotography_Milan_2026. The repository holds the five pipeline scripts, the final images and figures, and a README with setup instructions. With the seven FITS files in place, these commands rebuild every image, every figure and this PDF:

```
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt
./run_all.sh
```

The run takes about two minutes. Every parameter is set in one file, `pipeline/config.py`, and all of them are listed in the appendix. The software versions are in the next table.

Software	Version	Use
Python	3.14.4	pipeline language
NumPy	2.5.3	array maths
SciPy	1.18.1	Gaussian and median filters, FFTs, morphology
Astropy	8.0.1	FITS input and output, header checks
scikit-image	0.26.0	registration check, non-local-means denoising
Matplotlib	3.11.2	figures in this report
Pillow	12.3.0	PNG, JPEG and 16-bit TIFF export
WeasyPrint	70.0	this PDF

Credit

X-ray data: NASA / CXC, Chandra X-ray Observatory (chandra.harvard.edu). The data were not captured by Cepheid or by the authors. As the competition rules ask, this image must not be shared without this credit.

References

- Data: NASA / Chandra X-ray Center, chandra.harvard.edu. Not captured by Cepheid or by us.
- Ebeling, White & Rangarajan 2006, MNRAS 368, 65 (adaptive smoothing)
- Lupton et al. 2004, PASP 116, 133 (colour composites)
- Buades, Coll & Morel 2005, CVPR (non-local means)
- Strickland & Heckman 2009, ApJ 697, 2030 (M82 superwind)
- Lehnert, Heckman & Weaver 1999, ApJ 523, 575 (the cap)
- Bachetti et al. 2014, Nature 514, 202 (M82 X-2)

Appendix: parameters

Name	Value
BANDS	[(350, 500), (500, 700), (700, 1100), (1100, 1600), (1600, 2200), (2200, 2800), (2800, 6000)]
BKG_BIN	16
BKG_FIT_R	1200
BKG_MEDIAN_BOX	5
BLACK_NSIG	0.5
COLOUR_BLUR	5.0
CORE_HALF	420
DETAIL_KERNEL	2.0
DISTANCE_MPC	3.6
EXPOSURE_BAND	6
GALAXY_MASK_R	480
GROUP_COLOURS	{'R': (1.0, 0.42, 0.1), 'G': (0.3, 1.0, 0.35), 'B': (0.15, 0.4, 1.0)}
HIGHLIGHT_START	0.82
LOCAL_CONTRAST	(40.0, 0.35)
MIN_REL_EXPOSURE	0.2
NLM	{'patch_size': 5, 'patch_distance': 7, 'h_factor': 0.9}
PIXEL_ARCSEC	0.492
RGB_GROUPS	{'R': [0, 1, 2], 'G': [3, 4], 'B': [5, 6]}
SATURATION	1.45
SAT_RAMP	(0.08, 0.35)
SCALEMAP_MEDIAN	5
SCALEMAP_SMOOTH	1.5
SCALES	[0.5, 0.7, 1.0, 1.4, 2.0, 2.8, 4.0, 5.6, 8.0, 11.0, 16.0, 22.0, 32.0]
SHARPEN	(1.2, 0.4)
SNR_COLOUR	15.0
SNR_LUM	6.0
SOFT_NSIG	{'wide': 1.1, 'core': 12.0}
WHITE_PCT	{'wide': 99.995, 'core': 99.98}
WIDE_BOX	(540, 2740, 620, 2820)